Open problems:
Schrödinger equation is equivalent to information conservation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Classical counterpart.** State and prove the classical analogue: Hamiltonian evolution is equivalent to preservation of the Shannon entropy and of "the definitional independence of degrees of freedom". (Brief, Sec. 4: "This parallels the classical case where Hamiltonian evolution is equivalent to the preservation of Shannon entropy (Liouville's theorem)".)

2. **Entropy conservation as determinism and reversibility.** The brief reads DR-INFO as "an information theoretic way to characterize determinism and reversibility". A precise definition of deterministic and reversible evolution, and a proof that it is equivalent to DR-INFO, are not given. (Brief, Sec. 1.)

3. **Propagator for time-dependent $H(t)$.** "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$" is assumed. The formalization proves existence for constant H only; for bounded $H(t)$ it is the linear ODE existence theorem, and for unbounded self-adjoint $H(t)$ it needs Kato-type hypotheses. The converse has the same gap: for unbounded $H(t)$, Step 3 gives only a symmetric $H(t_0)$ (next section, first item). (Brief, Sec. 3, proof of Theorem 2 ($\Rightarrow$); Log, "Differences from the brief".)

From the formalization

Beyond finite dimension the formalization has a part in any dimension; it leaves open the items "Unbounded time-dependent Hamiltonians", "Equation at every time in infinite dimension", "Norm-differentiable $U(t)$ in infinite dimension" and "Wold–Cooper reduction" below.

1. **Unbounded time-dependent Hamiltonians.** EVDIFF on every density operator forces a bounded H (Banach–Steinhaus). With EVDIFF on a dense set of states, Theorem 2 is formalized for a time-independent evolution, with unbounded self-adjoint H (`drSceq_iff_drInfo_and_evDiff`, by Stone's theorem). Without EVTI, Step 3 gives only a skew-symmetric $A(t_0)$, and $H(t_0)$ need not be self-adjoint: the propagator of $H(t) = H_\beta$ for $t < 0$ and H_α for $t \geq 0$, two self-adjoint extensions of the momentum on $L^2(0,1)$, is differentiable at 0 exactly on the domain of the minimal momentum operator, which is not self-adjoint. Open: the hypotheses (Kato-type, or on the evolution) under which DR-INFO + EVDIFF gives a self-adjoint $H(t)$, and the formalization of the example. (Log, "Unbounded Hamiltonians".)

2. **Equation at every time in infinite dimension.** In finite dimension, one differentiable unitary $U(t)$ for all times solves the Schrödinger equation at every time, and $\Phi_{t_0 \to t_1}$ is conjugation by $U(t_1)U(t_0)^\dagger$ (`TwoParameter.drSceq_iff_drInfo_and_evDiff`). The proof fixes the determinant of the local unitaries, which leaves only a root of unity to match when gluing them. In any dimension the determinant is not defined, and the local unitaries are only strongly differentiable (next item), so the global propagator is not built there. (Log, "The equation at every time".)

3. **Norm-differentiable $U(t)$ in infinite dimension.** In any dimension only the strong equation $\psi'(t_0) = (i\hbar)^{-1} H\psi(t_0)$ is proved; the phase-fixing basis sum does not converge in operator norm. Whether a phase choice makes $t \mapsto U(t)$ differentiable in operator norm is open. (Log, any-dimension part, "Differences from the brief", EVDIFF item.)

4. **Wold–Cooper reduction.** The right shifts V_t on $L^2(0, \infty)$ give an evolution that satisfies [B]EVMIX, maps pure states to pure states, preserves transition probabilities and the entropy, but is induced by no unitary; this is formalized (`shift_semigroup_counterexample`, in the brief "Multiple equivalent characterizations of the Schrödinger equation"). The Wold–Cooper decomposition, which reduces every strongly continuous semigroup of isometries to a unitary group plus copies of this one, is not formalized. (Log, any-dimension part, "Problems found in the brief", second item.)